Customer Review Analysis: Clustering + Sentiment Classification

This project applies unsupervised **topic clustering** and **sentiment classification** to a corpus of 550 customer reviews spanning five product categories (Electronics, Clothing, Restaurant, Home Appliances, Beauty).

Dataset

550 customer reviews were used, spanning 5 categories (110 reviews each): **Electronics, Clothing, Restaurant, Home Appliances, Beauty**.

Example raw reviews:

- "Soft material and perfect fit, this pair of jeans is now my favorite."
- "This vacuum cleaner has made my daily chores so much easier, highly recommend."
- "Leaks water everywhere, this washing machine was a complete waste of money."

Preprocessing Pipeline

Methodology

Every review passes through the same deterministic pipeline before being vectorized:

1. **Lowercase** all text
2. **Strip noise** — URLs, HTML tags, emails, punctuation, and digits
3. **Tokenize** into individual words
4. **Remove stopwords** — standard English stopwords plus domain-specific low-signal words (e.g., "item", "product", "got")
5. **Lemmatize** each remaining token to its dictionary root form (WordNet lemmatizer), e.g. "amazing" stays, "lasts" → "last"
6. **Rejoin** into a cleaned string for TF-IDF vectorization

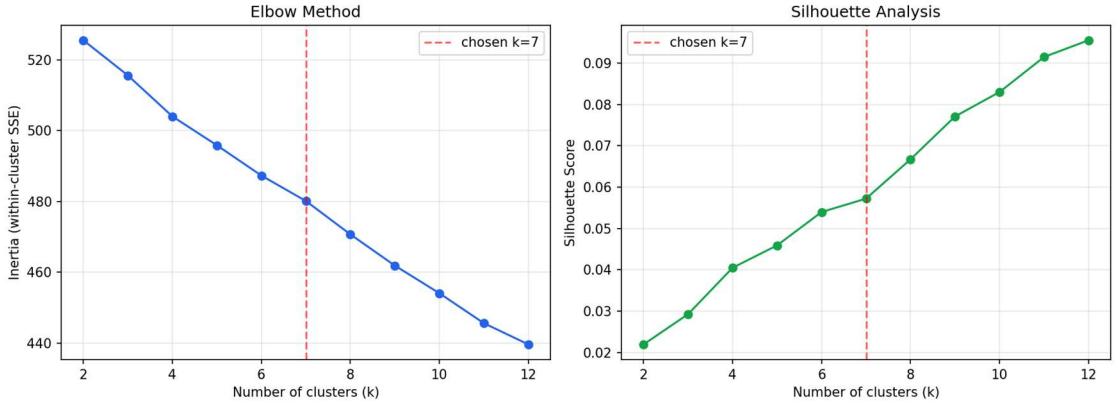
Example transformation (actual pipeline output)

Stage	Text
Raw	`This LAPTOP is AMAZING!!! Battery lasts 10+ hours, check http://example.com :)`
Cleaned	`laptop amazing battery last hour check`
Tokens	`['laptop', 'amazing', 'battery', 'last', 'hour', 'check']`

Methodology

1. Cleaned review text → **TF-IDF vectorization** (unigrams + bigrams, vocabulary capped at 2,000 terms, terms appearing in <3 or >85% of documents filtered out) → 550 × 797 feature matrix.
2. Fit **K-Means** across k = 2 to 12, tracking inertia (within-cluster sum of squares) and silhouette score at each k.
3. Selected **k via the elbow (knee) point** of the inertia curve rather than maximum silhouette — on sparse high-dimensional TF-IDF vectors, silhouette score tends to rise almost monotonically with k (favoring many tiny, overly specific clusters), so the elbow method gives a more interpretable and stable cluster count.
4. Fit the final K-Means model at the chosen k, extracted top TF-IDF terms per cluster (the cluster's "fingerprint"), and projected the feature space to 2D via PCA for visualization.

Choosing k



Elbow and silhouette analysis

The elbow method selected **k = 7**, at the point where inertia reduction visibly slows down (red dashed line).

Sample Cluster interpretation

Cluster	Size	Top terms	Dominant category	Sentiment mix (pos / neu / neg)
0	65	nothing, average, fine, special, work fine	Mixed (Beauty/Clothing/Restaurant)	28% / 12% / **60%**
1	30	money, waste, leak, water everywhere	**Home Appliances** (23/30)	0% / 0% / **100%**
2	86	decent, price, reliable, setup, performance	**Electronics** (53/86)	55% / 34% / 12%
3	32	exceeded expectation,	**Home Appliances** (23/32)	**100%** / 0% /

Cluster	Size	Top terms	Dominant category	Sentiment mix (pos / neu / neg)
		quiet operation, clean effectively		0%
4	71	pair, color, fit, fabric, jean, dress, wash	**Clothing** (71/71, pure)	58% / 8% / 34%
5	48	skin, irritation, lip balm, redness	**Beauty** (48/48, pure)	60% / 8% / 31%
6	218	use, arrived, sushi, platter, great, amazing	Mixed, Restaurant-leaning (84/218)	55% / 23% / 23%

Reading the clusters:

- **Clusters 1 and 3** are the most actionable finding: K-Means found a *purely negative* Home Appliances complaint cluster (leaking, "waste of money") completely separated from a *purely positive* Home Appliances praise cluster (quiet, efficient operation) — the clustering stage picked up on sentiment-driven vocabulary differences within the same product category without being told about sentiment at all.
- **Clusters 4 and 5** are clean, single-category clusters (Clothing and Beauty respectively), showing that fit/fabric language and skin/irritation language are each distinctive enough vocabularies to form their own well-separated topic.
- **Clusters 0 and 6** are cross-category "catch-all" clusters driven more by generic review phrasing ("nothing special," "arrived," "great," "amazing") than by product-specific vocabulary — a common and expected outcome in review clustering, since a meaningful fraction of review language is generic regardless of what's being reviewed.